

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	29 September 2026
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Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

Technical Architecture:

The deliverable includes the architectural diagram below and the information in Table-1 and Table-2.

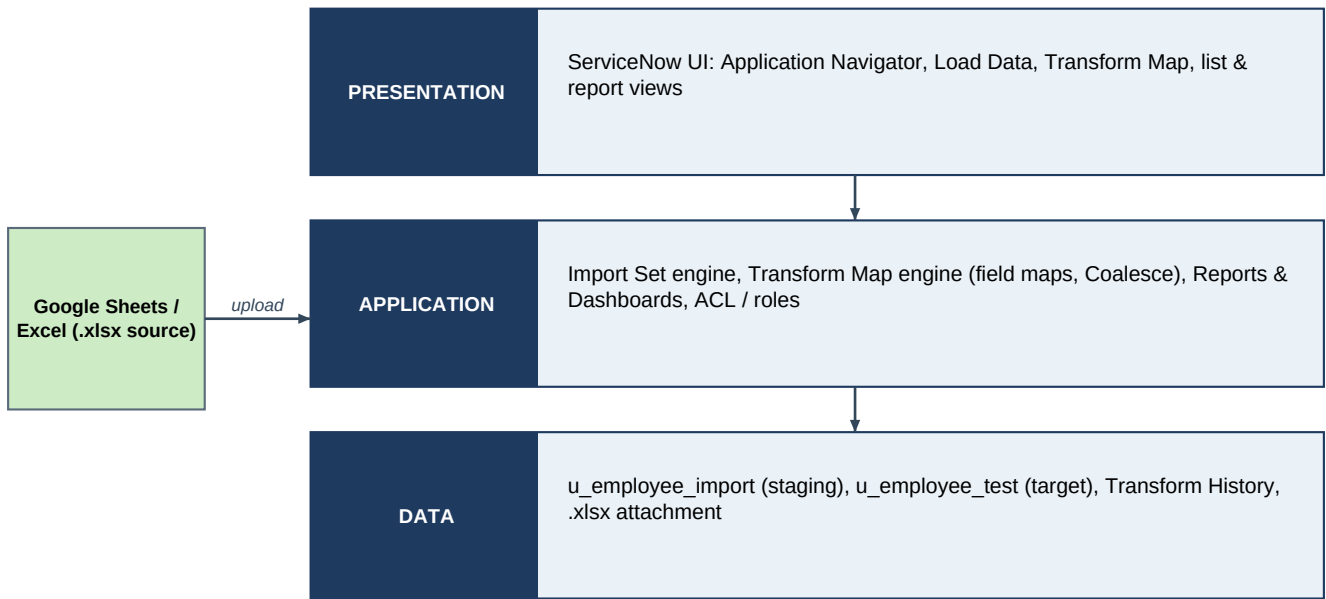


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Administrator works in the ServiceNow web UI: Application Navigator, Load Data, Transform Map, list and report views	ServiceNow Now Platform UI
2	Application Logic-1	Loads the uploaded spreadsheet into the staging table	Import Sets / Load Data (System Import Sets)
3	Application Logic-2	Maps and converts staging fields to the target table, with Coalesce on Employee ID	Transform Map with field maps and Coalesce
4	Application Logic-3	Builds reports and the Employee Analytics Dashboards dashboard	ServiceNow Reports and Dashboards (PA_DASHBOARDS)
5	Database	Staging table u_employee_import and target table u_employee_test; five String fields each	ServiceNow platform tables
6	Cloud Database	Instance database hosting the tables and Transform History	ServiceNow cloud instance (PDI)
7	File Storage	Excel file (Sample Spreadsheet.xlsx) attached during Load Data	ServiceNow attachments
8	External API-1	Source data prepared in a spreadsheet tool and downloaded as .xlsx	Google Sheets / Microsoft Excel
9	External API-2	Not used in this project	N/A

S.No	Component	Description	Technology
10	Machine Learning Model	Not applicable	N/A
11	Infrastructure (Server / Cloud)	Application runs on a ServiceNow Personal Developer Instance or an instance with development access	ServiceNow cloud

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	None - the solution is built with platform-native configuration, no custom code	ServiceNow low-code configuration
2	Security Implementations	Admin/development access required; ACL on pa_dashboards (Admin Overrides); report sharing restricted by users, groups and roles	ServiceNow ACLs and roles
3	Scalable Architecture	Staging table separates raw data from live records; same transform map reused for every upload	Import Set + Transform Map pattern
4	Availability	Hosted, managed ServiceNow instance; imports can be re-run at any time	ServiceNow cloud
5	Performance	Set-based transform of staged rows; unchanged rows are ignored via Coalesce; results in Transform History	Coalesce, Transform History

References:

ServiceNow Docs - Coalesce (Import Sets):

https://www.servicenow.com/docs/r/integrate-applications/system-import-sets/c_ImportSetCoalesce.html

ServiceNow Docs - Import Sets API (staging table, data source, transform map):

<https://www.servicenow.com/docs/r/application-development/servicenow-sdk/fluent-import-sets-api.html>

ServiceNow Docs - Import Sets key concepts: https://docs.servicenow.com/en-US/bundle/tokyo-platform-administration/page/administer/import-sets/concept/c_ImportSetsKeyConcepts.html